English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics / SEM, 24 August 2025 Examination problems — *The exact grading formula will depend on the results. As an approximation, use 2 plus the number of points. Time allowed: 3 hours. Good luck. You may use* $\sum_{k\geq 1} \frac{x^k}{k} = -\ln(1-x)$ *for* $|x| < 1$.

Problem 1. 1. (0.25 points) An urn contains n balls, and all possible numbers of white balls are equally likely. We add one white ball to the urn. What is the probability that a randomly selected ball from the urn is white?

2. (0.25 points) An urn contains n white and m black balls. Two players draw alternately from the urn, replacing each drawn ball. The first to draw a white ball wins. What is the probability that the first player wins? What happens as $m \to \infty$?

3. (0.25 points) We have two urns, 5 white balls and 5 black balls. How should we distribute the balls by number and colour between the urns to maximize the probability of drawing a white ball when we choose a random urn and then a random ball from it?

4. (0.5 points) Ivan has only 10-stotinka coins. The price of every item in a shop, in stotinki, ends in a random digit from $\{0, 1, 2, \ldots, 9\}$, with all final digits equally likely and independent. Ivan buys $n \geq 1$ items. What is the probability that he can pay exactly, without receiving change[1]? What is the expected number of items he must take until he can pay exactly for the first time[2]?

Problem 2. Consider a Bernoulli scheme (n, p). Let μ_n be the number of successful trials. The number $\tilde{p}_n := \mu_n/n$ is called the relative frequency of successes.

1. (0.25 points) If $p = 0.4$ and $n = 1500$, what is the probability that $\tilde{p}_n$ lies in $(0.4, 0.44)$?

2. (0.25 points) If $p = 0.375$, how many independent trials are needed so that $\{|\tilde{p}_n - p| \leq 0.01\}$ has probability at least 0.995?

3. (0.25 points) Let $p = 2/3$ and $n = 800$. What ε should we choose so that $\{|\tilde{p}_n - p| < \varepsilon\}$ has probability at least 0.985?

4. (0.25 points) If $n = 10000$, for what value of p will the probability of $\{|\tilde{p}_n - p| \leq 0.01\}$ be at least 0.99?

Problem 3. Let $p_k = ba^k/k$ for $k \geq 1$ and $0 < a < 1$.

1. (0.25 points) For what value of b is $(p_k)_{k\geq 1}$ a probability distribution?

2. (0.25 points) Find its probability generating function, expectation and variance.

3. (0.25 points) Let X have the distribution above. Find the probability generating function of $2X + 1$.

Problem 4. 1. (0.75 points) Three random points are chosen independently and uniformly on a circle of radius 1. Calculate the probability that the length L of the longest side of the triangle with these vertices does not exceed $\sqrt{3}$.

2. (0.25 points) Consider a generalization: how can we calculate $\mathbb{P}(L < \varepsilon)$, where $\varepsilon \leq \sqrt{3}$?

Problem 5. Let $U \sim U(0, 1)$ and $V \sim U(-\pi/2, \pi/2)$.

1. (0.25 points) One way to generate a pseudorandom number between 0 and 1 using U is to choose a natural number M, usually large, and calculate the fractional part of MU: $X := \{MU\}$. Find $\mathbb{E}X$, DX and $Cor(U, X)$. What do you think of this method?

2. (0.25 points) Let $U_1, \ldots, U_{100} \sim U$ be iid. What is the expectation of each? What about the largest and smallest among them?

3. (0.25 points) Find the density, expectation and variance of $Y := \tan V$.

4. (0.25 points) Let $Z \sim Cauchy(1)$, that is, $f_Z(x) = 1/(\pi(1 + x^2))$ for $x \in \mathbb{R}$. Prove that the random variables $1/Z$, $2Z/(1 - Z^2)$ and $(3Z - Z^3)/(1 - 3Z^2)$ have the same distribution.

[1] That is, his bill in stotinki is a multiple of 10.

[2] That is, the expected number of items until the total price in stotinki first becomes a multiple of 10.